

TAYLOR PENTZ

Senior Software Engineer

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Professional Summary

Senior Software Engineer with nearly a decade of experience designing and delivering large-scale software systems, including ML training pipelines, real-time data processing frameworks, and 3D scene understanding applications. Adept in architecting distributed infrastructure, leading development teams, and mentoring engineers to ensure technical excellence. Skilled at building visualization and verification tools to validate data integrity, training deep neural networks for computer vision tasks, and optimizing pipelines for performance in production environments. Proven ability to collaborate across functions and onboard engineers into high-velocity environments.

Technical Skills

- ❖ **Programming & Frameworks:** Python, TensorFlow, PyTorch, OpenCV, Keras, Scikit-learn, FastAPI, Flask, Node.js, Java, Scala
- ❖ **Cloud & DevOps:** AWS, GCP, Docker, Kubernetes, Terraform, CI/CD pipelines (GitHub Actions, Jenkins, GitLab CI)
- ❖ **Databases:** SQL (PostgreSQL, MySQL, MSSQL), NoSQL (MongoDB, DynamoDB, Redis, Cassandra)
- ❖ **Data & ML Engineering:** ML training pipelines, feature engineering, semantic segmentation, object detection, SLAM, 3D vision, depth estimation
- ❖ **Tools & Systems:** Airflow, MLflow, DVC, Ray, Kafka, Spark, Tableau, Grafana, Jupyter, Git
- ❖ **Practices:** Agile/Scrum, TDD/BDD, peer reviews, mentoring, recruitment & onboarding

Professional Experience

SumatoSoft

Senior Software Engineer

Boston, MA (Remote) | Apr 2023 – Mar 2025

- Designed and implemented production-ready ML training pipelines handling terabytes of multimodal data, ensuring reproducibility, scalability, and auditability.
- Built visualization and verification tools that validated 3D scene data quality and provided stakeholders with interactive insights for rapid decision-making.
- Trained and fine-tuned neural networks for semantic segmentation, object detection, and depth estimation, improving accuracy in 3D scene reconstruction tasks.
- Architected distributed ML infrastructure on AWS and GCP leveraging Docker and Kubernetes, streamlining deployment of AI models into real-time systems.
- Collaborated with research teams to adapt state-of-the-art SLAM algorithms for production environments, reducing latency and increasing stability of live systems.
- Mentored junior engineers, conducted code reviews, and standardized best practices across teams, raising overall code quality and reducing technical debt.

- Partnered with cross-functional stakeholders to translate business objectives into ML-driven features, accelerating delivery of roadmap goals and adoption.
- Recruited and onboarded engineers, facilitating smooth knowledge transfer and creating comprehensive onboarding guides to reduce ramp-up time.

Hexaview Technologies Inc.

Software Engineer

New York, NY | Jul 2018 – Mar 2023

- Implemented scalable ML data pipelines that ingested structured and unstructured data into SQL and NoSQL systems, enabling advanced AI applications.
- Contributed to computer vision research initiatives, prototyping deep learning models for object tracking and detection in dynamic real-time environments.
- Automated deployment of ML workloads with Docker and Kubernetes, ensuring robust and fault-tolerant operations across distributed systems.
- Led efforts to integrate cloud services (AWS Lambda, GCP BigQuery, Cloud Storage) into existing platforms, significantly enhancing scalability and throughput.
- Developed monitoring dashboards and visualization tools with Grafana and Tableau to provide transparency in model performance and system health.
- Coordinated sprint planning sessions with data scientists, product owners, and engineering leads to ensure accurate prioritization of ML initiatives.
- Improved CI/CD pipelines for ML workloads, introducing automated testing frameworks and model validation checkpoints prior to production release.
- Mentored junior developers in coding practices, DevOps workflows, and machine learning fundamentals, fostering professional growth and retention.

SimbirSoft Company

Software Developer

Boston, MA | Mar 2015 – Jun 2018

- Built backend services for AI-driven analytics platforms, integrating data from multiple sources and optimizing for high-throughput and reliability.
- Designed APIs to expose ML model predictions to downstream systems, ensuring low-latency responses for real-time data visualization applications.
- Created data pipelines for preprocessing and augmentation of computer vision datasets, enhancing performance of supervised deep learning models.
- Migrated critical services to containerized environments using Docker, simplifying deployment and reducing infrastructure maintenance overhead.
- Implemented testing frameworks for ML features, ensuring correctness and reproducibility of outputs across multiple environments and data sets.
- Collaborated with front-end teams to deliver interactive dashboards visualizing ML predictions, improving usability for non-technical stakeholders.
- Applied Agile practices in sprint cycles, enhancing team productivity and communication across international distributed teams.
- Participated in onboarding and pair programming sessions with new hires, accelerating their understanding of existing systems and workflows.

Microsoft

Software Developer Intern

Cambridge, MA | Aug 2014 – Feb 2015

- Assisted in building prototypes for computer vision research projects focused on depth estimation and 3D reconstruction using Kinect sensor data.
- Contributed to preprocessing pipelines for image and video datasets, ensuring consistency and quality for training experimental neural networks.
- Wrote automation scripts to streamline model training experiments and reduce manual intervention during iterative testing cycles.
- Supported senior engineers in debugging and optimizing code for experimental ML models running on GPU clusters.
- Developed internal visualization dashboards that presented intermediate model results to researchers, improving interpretability of ML outcomes.
- Collaborated with cross-disciplinary research teams to translate experimental concepts into implementable prototypes.
- Documented system architectures, experimental parameters, and outcomes, ensuring reproducibility for future research iterations.
- Participated in weekly team meetings and contributed to brainstorming sessions for enhancing project pipelines and tools.

Projects

- ✓ **3D Scene Understanding Pipeline:** Developed an end-to-end pipeline for semantic segmentation, object detection, and depth estimation in dynamic 3D environments.
- ✓ **SLAM Optimization:** Adapted and productionized state-of-the-art SLAM algorithms for real-time navigation in robotics systems.
- ✓ **Model Monitoring Dashboard:** Built interactive dashboards to visualize and validate ML model predictions, improving transparency and usability for stakeholders.

Certifications

- ❖ AWS Certified Solutions Architect – Associate
- ❖ Google Cloud Professional Machine Learning Engineer
- ❖ Docker Certified Associate

Education

University of North Carolina at Chapel Hill

Bachelor's Degree in Computer Science (2010 – 2014)